



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECP3213: BUSINESS INTELLIGENCE

2025/2026 – SEMESTER 2

PROJECT PHASE 2:

Phase 2: Data Workflow (Alteryx Designer)

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1.0 Workflow Overview & Tools Used

To design the Business Intelligence architecture, our team utilized Alteryx Designer to construct our complete workflow.

Alteryx Tools utilized:

- **Input/Output:** Data Input, Data Output
- **Data Preparation (Cleaning):** Filter, Data Cleansing, Unique, Select, Sort
- **Data Transformation & Integration:** Join, Formula, Summarize
- **Data Investigation (Analytical Techniques):** Pearson Correlation

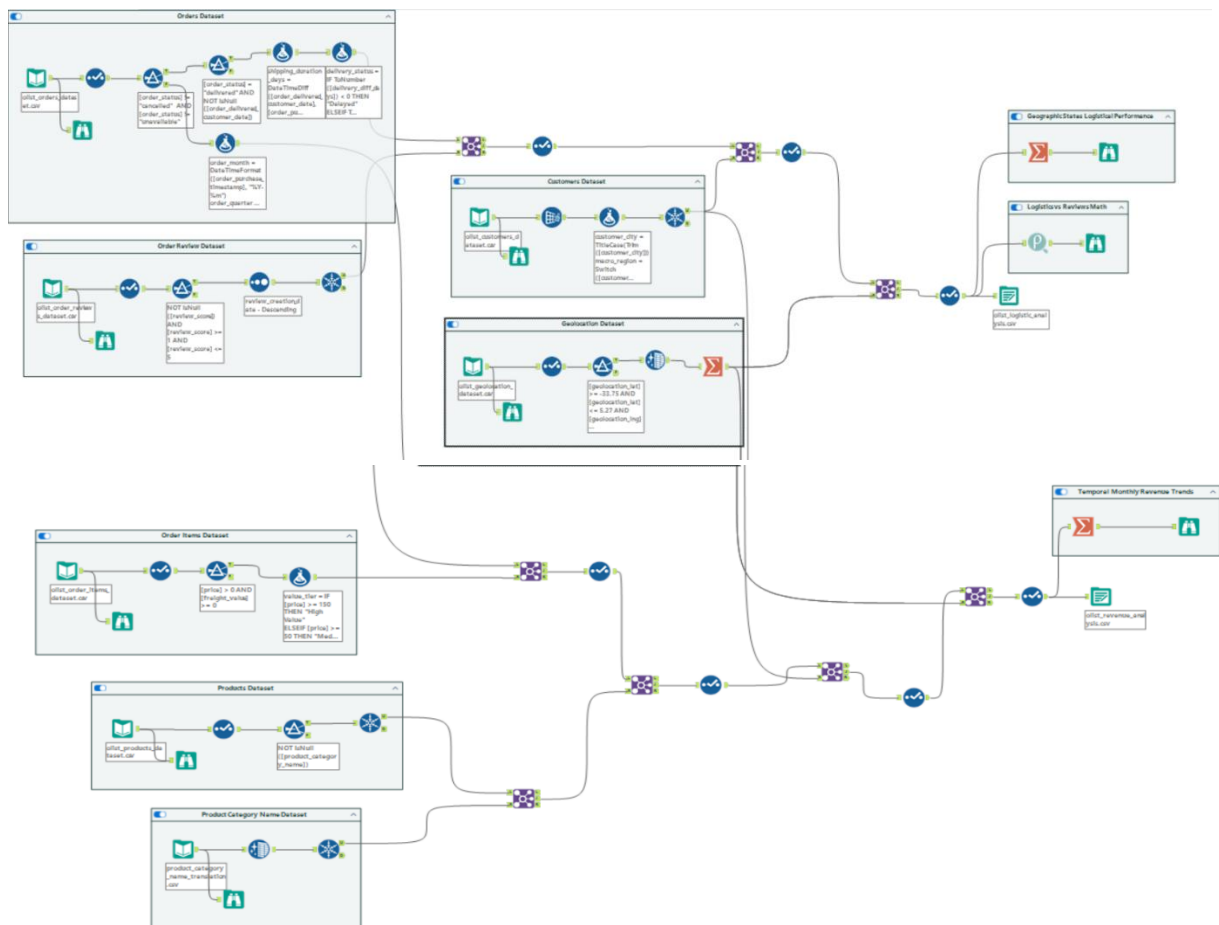
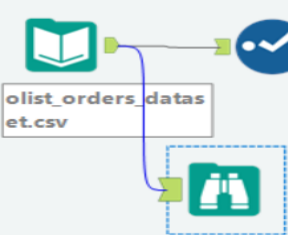


Figure 1: Workflow Overview

2.0 Data Input from Multiple Datasets

Our workflow ingested seven distinct CSV datasets from the Olist Brazilian E-Commerce database. This was executed using multiple Input Data tools configured to read comma-delimited files.

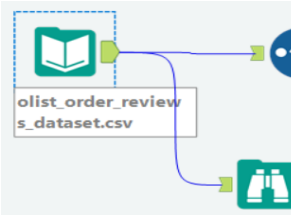
1. **Orders Dataset (olist_orders_dataset.csv):** Contains core transaction statuses and purchase timestamps.



Record	Name	Type	Size
1	order_id	V_String	254
2	customer_id	V_String	254
3	order_status	V_String	254
4	order_purchase_timestamp	V_String	254
5	order_approved_at	V_String	254
6	order_delivered_carrier_date	V_String	254
7	order_delivered_customer_date	V_String	254
8	order_estimated_delivery_date	V_String	254

Figure 2.1: olist_orders_dataset.csv

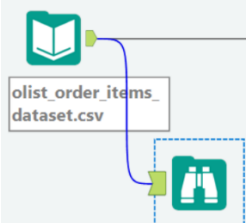
2. **Order Reviews Dataset (olist_order_reviews_dataset.csv):** Captures post-purchase customer feedback, specifically the 1-to-5 satisfaction scores and timestamped review dates.



Record	Name	Type	Size
1	review_id	V_String	254
2	order_id	V_String	254
3	review_score	V_String	254
4	review_comment_title	V_String	254
5	review_comment_message	V_String	254
6	review_creation_date	V_String	254
7	review_answer_timestamp	V_String	254

Figure 2.2: olist_order_items_dataset.csv

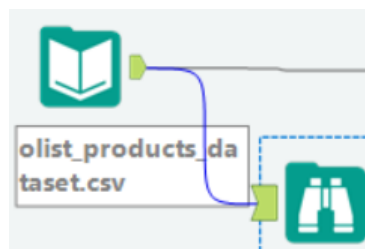
3. **Order Items Dataset (olist_order_items_dataset.csv):** Contains the line-item financial details for each transaction, including product IDs, item prices, and freight values.



Record	Name	Type	Size
1	order_id	V_String	254
2	order_item_id	V_String	254
3	product_id	V_String	254
4	seller_id	V_String	254
5	shipping_limit_date	V_String	254
6	price	V_String	254
7	freight_value	V_String	254

Figure 2.3: olist_order_items_dataset.csv

4. **Products Dataset (olist_products_dataset.csv):** Contains physical product attributes and assigns a Portuguese product category name to each item ID.



Record	Name	Type	Size
1	product_id	V_String	254
2	product_category_name	V_String	254
3	product_name_lenght	V_String	254
4	product_description_lenght	V_String	254
5	product_photos_qty	V_String	254
6	product_weight_g	V_String	254
7	product_length_cm	V_String	254
8	product_height_cm	V_String	254
9	product_width_cm	V_String	254

Figure 2.4: olist_products_dataset.csv

5. **Product Category Name Translation (product_category_name_translation.csv):** A mapping table used to seamlessly translate the Portuguese product category names into English, ensuring the final dashboards are readable for English-speaking stakeholders.



Record	Name	Type	Size
1	product_category_name	V_WString	254
2	product_category_name_english	V_WString	254

Figure 2.5: product_category_name_translation.csv

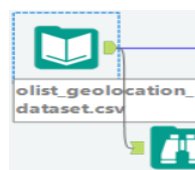
6. **Customers Dataset (olist_customers_dataset.csv):** Provides demographic mapping, linking unique customer IDs to their geographic locations which includes city, state, and zip code prefix.



Record	Name	Type	Size
1	customer_id	V_String	254
2	customer_unique_id	V_String	254
3	customer_zip_code_prefix	V_String	254
4	customer_city	V_String	254
5	customer_state	V_String	254

Figure 2.6: olist_customers_dataset.csv

7. **Geolocation Dataset (olist_geolocation_dataset.csv):** A highly granular spatial lookup table mapping Brazilian zip codes to specific latitude and longitude coordinates.



Record	Name	Type	Size
1	geolocation_zip_code_prefix	V_String	254
2	geolocation_lat	V_String	254
3	geolocation_lng	V_String	254
4	geolocation_city	V_String	254
5	geolocation_state	V_String	254

Figure 2.7: olist_geolocation_dataset.csv

2.1 Data cleaning (missing values, duplicates)

2.1.1 Select Tool configuration

Column	Type	Size	Rename	Description
order_id	V_String	254		
customer_id	V_String	254		
order_status	V_String	254		
order_purchase_timestamp	DateTime	19		
order_approved_at	DateTime	19		
order_delivered_carrier_date	DateTime	19		
order_delivered_customer_date	DateTime	19		
order_estimated_delivery_date	DateTime	19		
*Unknown	Unknown	0		Dynamic

A Select Tool was applied immediately after the Input Data tool to correct the data types of all five timestamp columns from V_String to DateTime. This conversion is essential as Alteryx cannot perform date difference calculations or time-series operations on string-formatted date values.

2.1.2 Filter Tool configuration

```
[order_status] != "cancelled" AND [order_status] != "unavailable"
```

Orders Dataset

Disable container

olist_orders_datas et.csv

[order_status] != "cancelled" AND [order_status] != "unavailable"

[order_status] = "delivered" AND NOT IsNull ([order_delivered_customer_date])

shipping_duration = DateTimeDiff ([order_delivered_customer_date], [order_purchase_timestamp], "days")

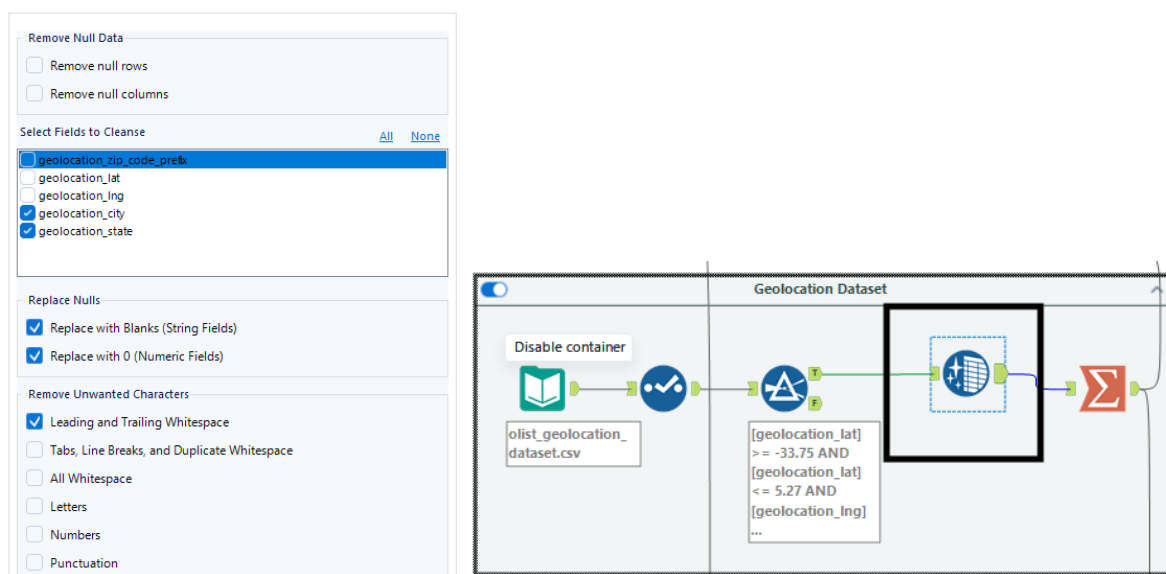
delivery_status = IF ToNumber ([delivery_diff_days]) < 0 THEN "Delayed" ELSEIF T...

order_month = DateTimeFormat ([order_purchase_timestamp], "%Y-%m")

order_quarter ...

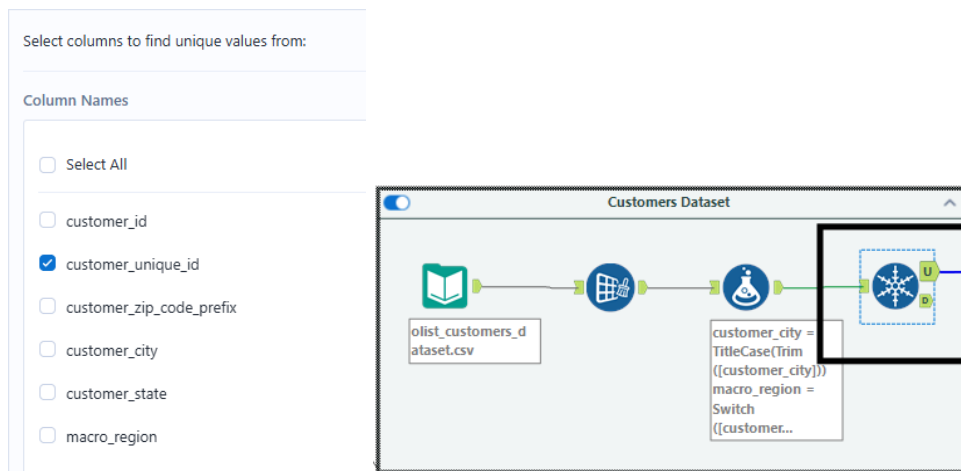
A second Filter Tool was applied to split the cleaned orders dataset into two analysis streams. The True output produces Stream A containing only fully delivered orders with confirmed customer delivery dates, which is used for logistics performance and customer satisfaction analysis. The False output produces Stream B containing all remaining non-cancelled orders regardless of delivery status, which is used for revenue trend and product category analysis. This data split customizes the input for each analytical objective while maintaining complete data integrity across both paths.

2.1.3 Data Cleanse Tool configuration



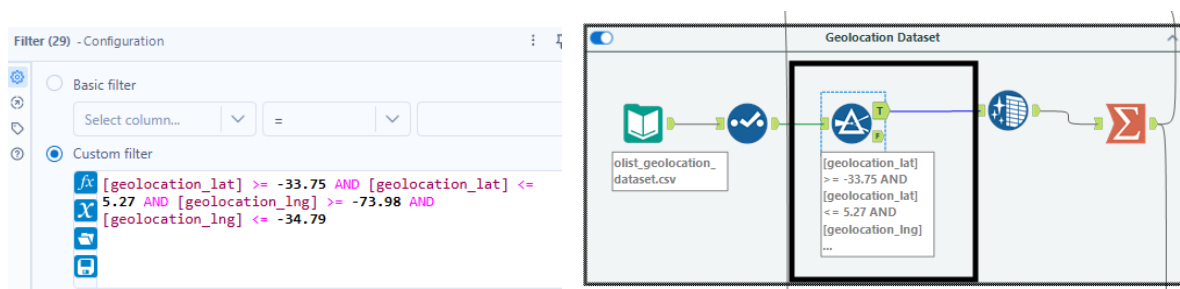
A Data Cleanse Tool was applied to the geolocation dataset to remove leading and trailing whitespace from the `geolocation_city` and `geolocation_state` columns. Inconsistent whitespace in geographic fields would cause join failures and incorrect grouping during aggregation, as entries such as 'sao paulo' and 'sao paulo' would be treated as two distinct cities.

2.1.4 Unique Tool configuration



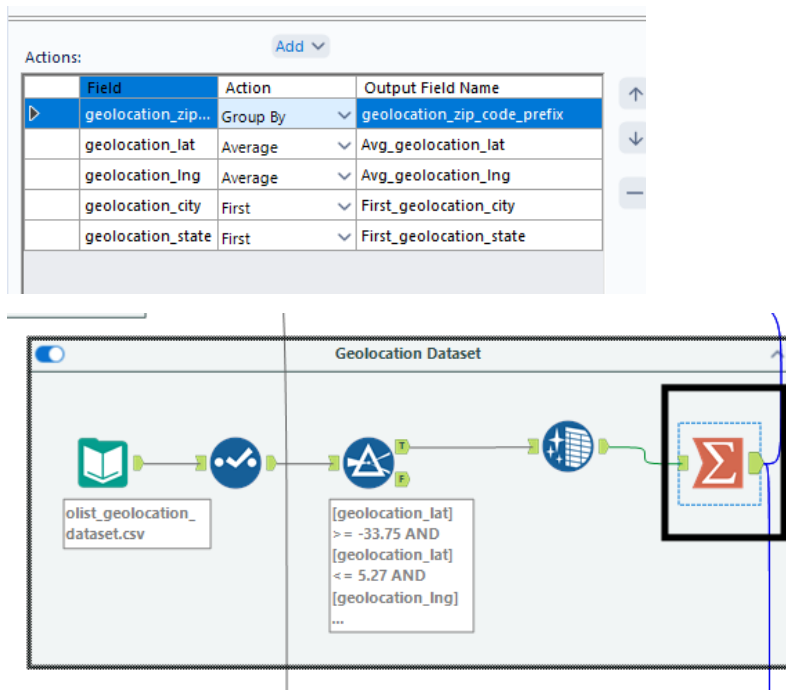
A Unique Tool was applied to the customers dataset to remove duplicate customer records based on the customer_unique_id field. In the Olist dataset, a single customer may have placed multiple orders, each generating a separate customer_id entry. However, the customer_unique_id field represents the actual individual customer regardless of how many orders they placed. Deduplicating on this field ensures that customer counts in geographic segmentation analysis reflect the true number of unique individuals rather than the total number of orders placed.

2.1.5 Filter Tool configuration (geolocation dataset)



A Filter Tool was applied to the geolocation dataset to remove coordinate entries that fall outside Brazil's valid geographic boundaries. The filter expression retains only records where geolocation_lat is between -33.75 and 5.27 and geolocation_lng is between -73.98 and -34.79, which represent the southern and northern extremities of Brazil respectively.

2.1.6 Summarize Tool configuration (geolocation dataset)



The geolocation dataset contains many duplicate zip code entries with slightly different coordinate values. To handle this, a Summarize Tool was used to collapse all duplicate zip codes into one single row per zip code. For each zip code, the latitude and longitude values are averaged across all duplicate entries to produce a more accurate central coordinate point. The city and state values are kept using the First method. The final result is one clean row per zip code, ready to be joined with the customers dataset.

2.2 Data transformation (join, filter, aggregation)

2.2.1 Formula Tool configuration (

2.3 Data integration (combine datasets)

2.4 Feature creation (if applicable)

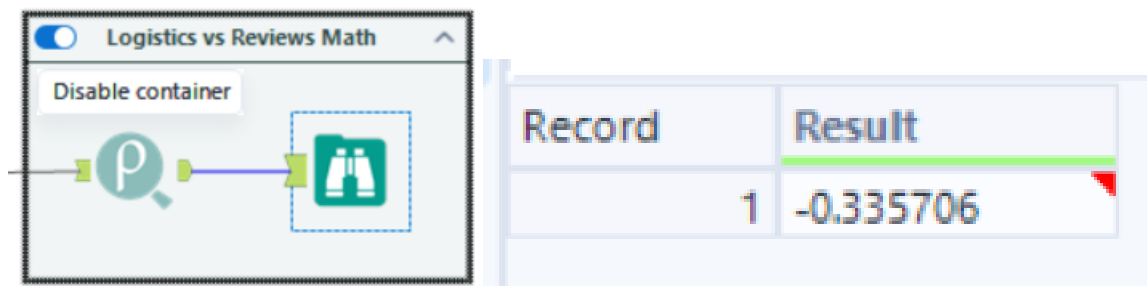
3.0 Analytical Techniques & Results

We embedded three core analytical methods to extract business intelligence value before exporting to destination files:

3.1 Predictive Logic (Pearson Correlation Analysis)

Used a **Pearson Correlation Tool** to test how shipping intervals affect customer brand loyalty, the completed logistics data stream was routed.

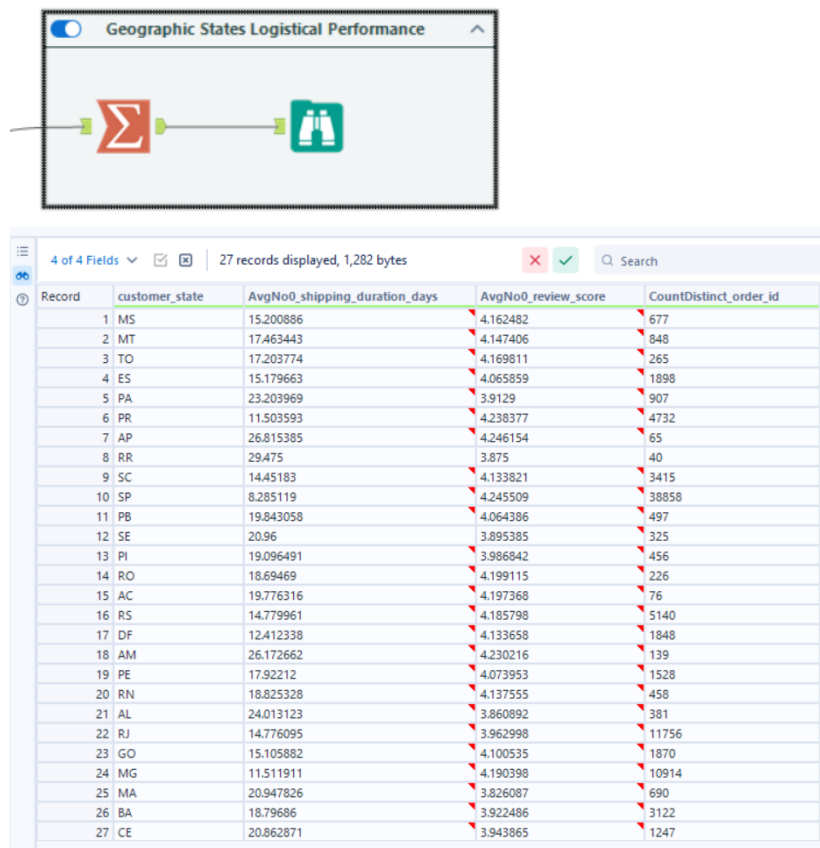
Configured Variables	shipping_duration_days and review_score.
Empirical Finding	The engine yielded an exact correlation coefficient (\$r\$) of - 0.3357 .
Business Evaluation	This proves a moderate negative linear correlation . It provides mathematical proof that as shipping times lengthen, customer review scores drop predictably, establishing an empirical baseline for logistics optimization.



3.2 Geographic Segmentation Matrix

A **Summarize Tool** was attached to the Logistics Stream to group operational performance across regional segments.

Configuration	Grouped by customer_state, aggregating Average(shipping_duration_days) and Average(review_score).
Result	Outputs a structural matrix evaluating logistics bottlenecks and satisfaction grades across all 27 Brazilian regions.



3.3 Temporal Trend Analysis Matrix

A secondary **Summarize Tool** was linked to the **Revenue Stream** to evaluate historical business growth velocity over time.

Configuration	Grouped by the custom order_month field, calculating Sum(payment_value) and Count Distinct(order_id).
Result	Aggregates a time-series matrix measuring transactional demand and commercial velocity across Olist's operating history.

